

KURT RIGGIN

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EDUCATION

Master of Science in Biostatistics (Capstone Track)

University of Washington, Sep 2026 - Present

- Expected graduation: March 2028

Bachelor of Science in Biology, Minor in Mathematics

Regis University, Aug 2022 - Dec 2025

- Cumulative GPA: 4.0
- Honors Program; Undergraduate Honors Thesis - Influence of Visual Communication on Scientific Literacy
- Relevant Coursework - Environmental Biostatistics/Research Design (graduate-level independent study), Bayesian Data Analysis (independent study), Calculus-Based Statistics (independent study), Linear Algebra, Calculus I-III

RESEARCH EXPERIENCE

Colorado Summer Institute in Biostatistics (CoSIBS)

University of Colorado Anschutz, Jun 2023 - Aug 2023

- Developed corradar, an R-based data visualization improvement of normal radar plots using a correlation-based system to determine axes' positions, under Dr. Carsten Görg & Dr. Ryan Peterson.
- Tested corradar functionality using force-based algorithms with simulated annealing, k-means clustering, data plotting, and presenting work at daily team meetings.
- Completed biostatistical coursework and training in R within a selective cohort of 20 students.
- Applied biostatistical methods and corradar plots in a hackathon analyzing asthma respirator data.
- R package in development.

NRSA Rivers and Streams Undergraduate Research

Regis University, Jan 2025 - Present

- Cleaned and combined data from the 2018-2019 NRSA rivers and streams dataset, fish trait data (rfishbase R package), and phylogenetic trees inferred from GenBank.
- Ran joint species distribution models (Hmsc package in R) to estimate the *Catostomidae* distribution across the Northeastern United States, under Dr. Kristofor Voss.
- Fit Bayesian models and assessed MCMC convergence diagnostics.
- Produced R Markdown research reports documenting code, findings, and challenges.
- Expanding research to compare RIVPAC, SDM, and JDSM models of 2013-2014 NRSA rivers and streams fish assemblages in the Eastern Highlands.

Emberizoid Hindlimb Undergraduate Research

Regis University, Jan 2024 - Jul 2025

- Characterized the ecomorphology and phylogenetic signal of hindlimbs in North American sparrows and blackbirds (*Passerellidae* & *Icteridae*) under Dr. Tyler Imfeld.
- Collected anatomical data using precision calipers on 9 hindlimb traits from specimens provided by the Denver Museum of Nature & Science.
- Performed statistical analyses in R (phylogenetic and non-phylogenetic): PCA, ANOVA, GLMs, MANOVA, regression, and ancestral character estimation.
- Created PCA and violin plots for research presentations.
- Manuscript in preparation.

Acropoma Diet Partitioning Undergraduate Research

Regis University, Aug 2024 - Dec 2025

- Analyzed seasonal diet partitioning in *Acropoma banedai* & *Acropoma japonicum* under Dr. Mike Ghedotti & Dr. Kristofor Voss.

- Fit GLMMs with fixed and random effects to assess size and diet relationships, and PERMANOVA to calculate variance explained.
- Calculated Bray-Curtis dissimilarity matrices, index of relative importance, and niche breadth and overlap.
- Drafted statistical methods and results and generated segmented bar charts for publication.

Behavioral Forest Edge Undergraduate Research

Regis University, Aug 2025 - Present

- Investigated Depth of Edge Influence (DEI) for anthropogenic and river edge effects on the behavior of howler monkeys (*Alouatta palliata*) in Costa Rica under Dr. Amy Schreier and Dr. Kristofor Voss.
- Fit and compared multiple models (null, linear, power, logistic, segmented, stepwise) using AIC to assess model fit.
- Processed and analyzed behavioral data in R (tidyverse).
- Drafted statistical methods and results, with manuscript in preparation.

Cranial Photogrammetry Undergraduate Research

Regis University, Mar 2025 - Aug 2025

- Constructed a photogrammetry rig to create 3D scans of bird skulls provided by the Denver Museum of Nature & Science, under Dr. Tyler Imfeld.
- Accessed and generated models using the MorphoCloud online desktop instance, using the SlicerMorph Photogrammetry extension created by Dr. Murat Maga in 3DSlicer.
- Created instructional documentation to train students in MorphoCloud photogrammetry.

Deep Sea Histology Undergraduate Research

Regis University, Aug 2022 - Aug 2023

- Investigated bioluminescence in dorsolateral “white line” tissue of *Eurypharynx pelecyanoides* under Dr. Mike Ghedotti.
- Conducted histological analysis using paraffin embedding, rotary microtome sectioning, and staining (Toluidine Blue & Masson’s trichrome protocol).
- Examined and photographed histological sections to prepare images for publication.

OPEN-SOURCE SOFTWARE

standrecon R Package

Regis University, Aug 2025- Present

- Developed an R package to reconstruct forest stand conditions summed by tree density and basal area at any given reference year using contemporary data, under methods developed by Dr. John Sakulich.
- Currently developing a manuscript for testing the package on Denver Parks and Recreation for fire regime specifications.

PUBLICATIONS

Ghedotti, M. J., **Riggin, K. L.**, Marsh, M.A., Bramlett, S.A., Pearson, H.D., Rodriguez, G.J., Gruber, J.N., Egan, J.P., & Voss, K.A. (2026). Evidence of deep-sea trophic resource partitioning between the glowbellies *Acropoma hanedai* and *A. japonicum* near southwestern Taiwan. *Environmental Biology of Fishes*, 109, 43.

<https://doi.org/10.1007/s10641-026-01814-y>

Submitted Manuscripts in Review

Ghedotti, M. J., Gonzalez, F. M., **Riggin, K. L.**, Agudo, K. C., Davis, M. P., Smith, W. L., & Frable, B. W. (in review). Anatomy and Evolution of Bioluminescent Organs in the Gulper Eels (*Anguilliformes: Saccopharyngoidea*). *Ichthyology & Herpetology*.

WORK EXPERIENCE

Academic Tutor

Regis University, Aug 2025 - Dec 2025

- Tutored undergraduate students in biology, physics, chemistry, and mathematics.
- Provided instruction and tutoring in R programming for a quantitative ecology laboratory course.

Student Advancement Ambassador

Regis University, Aug 2022 - May 2025

- Raised money from alumni for scholarships to incoming university students.
- Recorded promotional and thank-you videos for donations to university funding.

PRESENTATIONS

Riggin, K.L. (presenter), Sicut, Z.J., & Imfeld, T.S. (2025). Ecomorphology and phylogenetic signal of hindlimbs in North American sparrows and blackbirds (*Passerellidae* & *Icteridae*). Colorado Field Ornithologists (CFO) – Grand Junction, CO – Jun. 5-8, 2025. Abstract published in *Colorado Birds* (Summer 2025). [Oral Presentation]

Riggin, K.L. (presenter), Sicut, Z.J., & Imfeld, T.S. (2024). Ecomorphology of the hindlimb in North American sparrows. American Ornithological Society (AOS) – Estes Park, CO – Oct. 1-5, 2024. [Poster Presentation]

Riggin, K.L. (presenter). (2025). Influence of visual communication on scientific literacy (Honors Thesis Defense). Regis University – Denver, CO – Dec. 5, 2025. [Oral Presentation]

Riggin, K.L., Heiser, G., Indeglia, K., Shim, J., Greb, N., Peterson, R., & Görg, C. (2023). Statistical analysis and exploration of data from patients with asthma with force-directed correlation-based radar plots. Poster presented at CoSIBS 2024 – Denver, CO – Jul. 25, 2023. [Poster Presentation]

Ghedotti, M. J. (presenter), K.C. Agudo, F.M. Gonzalez, **Riggin, K.L.**, Davis, M.P., Smith, W.L., & Frable, B.W. (2023). Anatomy and Evolution of Bioluminescence in the Gulper Eels (*Anguilliformes*). Society for Integrative and Comparative Biology (SICB) – Seattle, WA – Jan. 2-6, 2024. Abstract published in *Integrative and Comparative Biology*, 64. [Poster Presentation]

Ghedotti, M. J. (presenter), K.C. Agudo, F.M. Gonzalez, **Riggin, K.L.**, Davis, M.P., Smith, W.L., & Frable, B.W. (2023). Anatomy and Evolution of Bioluminescence and Ultra-Black Coloration in the Gulper Eels (*Anguilliformes*). American Society of Ichthyologists and Herpetologists (ASIH) – Norfolk, VA – July 12-17, 2023. [Poster Presentation]

SKILLS

Programming & Statistical Software:

- R: R Markdown, tidyverse, dplyr, ggplot2, Hmsc, vegan, nls

Statistical Methods:

- GLMMs, PCA, ANOVA, JSDMs, Regression

Other Software:

- 3D Slicer, SlicerMorph

AWARDS

Board of Trustees Scholarship

Regis University, Aug 2022 - Present

Science Distinction Award

Regis University, Aug 2022 - Present

Undergraduate Research (URISE) Travel Award

Regis University, Aug 2024

References available upon request.